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Applying Extreme Value Theory to a Century of Ammonium Nitrate Disasters: Persistent Safety Risks in Chemical Supply Chains

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ABSTRACT

We have analyzed 39 major ammonium nitrate (AN) accidents (1916–2022) that reveal systemic catastrophic risks have not diminished over the past century. This challenges the prevailing narrative of improved process safety of AN on account of iterative guidelines and technology advances. Applying extreme value theory (EVT), we model temporal occurrence as a homogeneous Poisson process (maximum likelihood estimate $\lambda \approx 0.23$ events/year) and casualty severity as heavy-tailed (generalized Pareto distribution, shape parameter $\xi \approx 1.13$). These findings indicate that there are persistent systemic patterns. This is shown by the implied "shadow mean" that exceeds the sample mean by 1.5–19-fold, and societal risk profiles occupy the "intolerable" region per frequency–number curve criteria. To address this problem, we establish an EVT-based systemic monitoring framework with quantitative benchmarks, identifying 50-year (~315 fatalities) and 100-year (~715 fatalities) return levels as a data-driven baseline for assessing safety performance. This ensemble framework provides a necessary tool for stakeholders to monitor realized systemic risks over mechanism-driven simulations or narrative expectations.

1 | Introduction

Ammonium nitrate (AN) is a dual-use compound (NH_4NO_3) produced in mass quantities for agriculture, explosives, and other applications. AN has been observed to have inherent explosive risks under conditions like heat, contamination, or confinement. Major accidents—such as Oppau (1921, 561 deaths), Texas City (1947, 581 deaths), Toulouse (2001, 31 deaths), and Beirut (2020, 218 deaths, with a $15B economic toll)—highlight persistent hazards across its global supply chain, from production to transport and storage (Al-Hajj et al. 2021).

For this study, we have employed a commonly used definition for major AN incidents, which are defined as those involving the explosion of more than 1,000 tons of AN or resulting in more than 30 fatalities, as classified in comprehensive historical reviews (CAS 2021; Pittman et al. 2014; Babrauskas 2016). This dataset comprises 39 events from 1916–2022, including 14 zero-fatality incidents that meet the mass threshold and represent significant systemic failures with high catastrophic potential (e.g., explosions with near-misses in supply chains). While these are included for a comprehensive historical record, our severity modeling focuses on the 25 fatal accidents to characterize the tail behavior of realized human risks.

We challenge the prevailing narrative that iterative guidelines and technological advances have improved AN process safety, as these events are not isolated anomalies but rather realizations from a complex, coupled system where stochastic interactions—

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between human factors, environmental variables, and material properties—can escalate local failures into catastrophic outcomes.

Process safety literature has framed these risks through two paradigms employing reductionist logic, which miss the key point of the role of randomness and luck, as they treat inherently stochastic, emergent phenomena as amenable to deterministic analysis. These approaches overlook systemic tail risk, defined as the occurrence of low probability but high impact events. The first relies on qualitative narratives emphasizing "lessons learned," which compile incident details into prescriptive guidelines, arguing that iterative regulations and technological advances progressively mitigate or prevent the worst hazardous outcomes (CAS 2021; Pittman et al. 2014; Babrauskas 2016). While these are appealing to do in the immediate aftermath of accidents to provide policy input, they extrapolate from individual events to systemic generalizations without rigorous statistical validation, overlooking randomness and coupling in complex systems.

The second paradigm is site-specific quantitative risk assessments (QRAs), which employs a bottom-up mechanistic viewpoint, integrating probabilistic failure rates (e.g., via event trees modeling barriers as binary outcomes with conditional probabilities) and deterministic consequence simulation models to map event-specific outcomes such as fires, explosion, and toxic dispersions (Sivaraman and Varadharajan 2021; CCPS 2007). Although widely used in industry for safety assessment and control, QRAs focus on component interactions but often fail to fully capture non-linear, unforeseen couplings that render some aspects of complex systems computationally irreducible. In complexity science, Wolfram (2017) posited that emergent behaviors arising from simple rules cannot be deterministically predicted via analytical shortcuts or predefined models. Instead, we argue that statistical modeling of historical ensembles offers a more appropriate method to analyze these stochastic phenomena, revealing systemic patterns overlooked by conventional approaches.

More importantly, neither paradigm aggregates data across incidents to test scientific claims of declining risks, as they cannot generalize from particular events (random accidents) to the ensemble's heavy-tailed dynamics at the system level. These inferences, which are not rigorously verified, can give the illusion of control and safety while underestimating tail risks in reality (Perrow 1984; Taleb 2020).

We address this gap by analyzing 39 major AN accidents (1916–2022) as a statistical ensemble, applying extreme value theory (EVT) to model risk patterns—inspired by Cirillo and Taleb (2016) on heavy-tailed conflict severities. We test whether AN accidents exhibit similar persistent heavy-tailed distributions, driven by a constant hazard rate and severe tails that reveal extreme casualties (>500 deaths) as a systemic feature. This ensemble approach quantifies realized systemic risks, offering a data-driven foundation inviting us to rethink safety frameworks beyond site-specific measures, with implications for global chemical risk management.

2 | Results

2.1 | The Rhythm of Risk: Temporal Analysis

We analyzed the temporal distribution of AN accidents, which reveals a consistent hazard rate. This challenges narratives of declining risk through regulatory or technological advances. We analyzed a dataset of 39 major AN incidents from 1916 to 2022 (shown in Figure 1), comprising 25 fatal accidents (casualties range from 1 to 581, mean 101.5 and median 20) and 14 zero-fatality events. The inclusion of non-fatal events provides a complete ensemble, representing near-misses that inform the Poisson process without affecting severity models.

The focus here is on the 25 fatal accidents, whose inter-arrival times, T_i (24 intervals, mean 4.36 years), which we modeled as a homogeneous Poisson process. The maximum likelihood estimate of the Poisson rate parameter $\lambda \approx 0.23$ events/year corresponds to the constant hazard rate of the exponential inter-arrival times (mean $\approx$ 4.36 years). This estimate, visualized in Figure 2a, shows an exponential fit to the inter-arrival distribution (PDF: $f(T) = 0.229e^{-0.229T}$), with the quantile–quantile (QQ) plot (Figure 2b) and ACF plot (Figure 2c) confirming the process's memoryless nature (ACF $\approx$ 0) and alignment with an exponential model. Bootstrap resampling with 10,000 iterations (Supplementary Figure S2.1) validated this λ, with the distribution centered around 0.23 events per year and 95% percentile CI [0.15, 0.39], indicating robustness despite the sample size ($n = 25$ fatal events). A dispersion test on event counts per decade further supported this constancy (a dispersion index of 1.064, a chi-squared statistic of 10.64, and a p-value of 0.38624). Moreover, this result confirms the suitability of Poisson model over more complex alternatives, such as the negative binomial model (Das and Dey 2016).

To assess the hazard's behavior for fatal risks, we conducted a peaks-over-threshold (POT) analysis, varying casualty thresholds from 0 to 500 deaths (Supplementary Figure S2.2). The initial $\lambda \approx 0.23$/year, derived from the 25 fatal accidents, decreased progressively, stabilizing around 0.05 events per year for thresholds exceeding 50–100 casualties. The reduction to 0.05/year is expected in a Poisson process, which means that severe events (e.g., >500) have lower λ because they are rarer (have longer return periods), not because the hazard decays. Sensitivity analysis via POT (Supplementary Figure S2.2) showed that λ's decrease is modest (10%–15% at high thresholds), validating its stability across varying severity levels and reinforcing the temporal constancy established by the bootstrap results (Supplementary Figure S2.1). This consistency aligns with the dataset's unchanging hazard rate over the 106-year record.

This persistent hazard rate suggests that AN risks remain unchanged over time, irrespective of interventions. Therefore the clusters (e.g., 1910s–1940s) and gaps (e.g., 1947–2001) observed in the dataset are natural features of a Poisson process, where variance equals the mean, allowing for apparent "clustering" or "quiet periods" as random fluctuations rather than evidence of temporal dependence or declining risk (as confirmed by ACF $\approx$ 0, Figure 2c). This aligns with stochastic luck in complex

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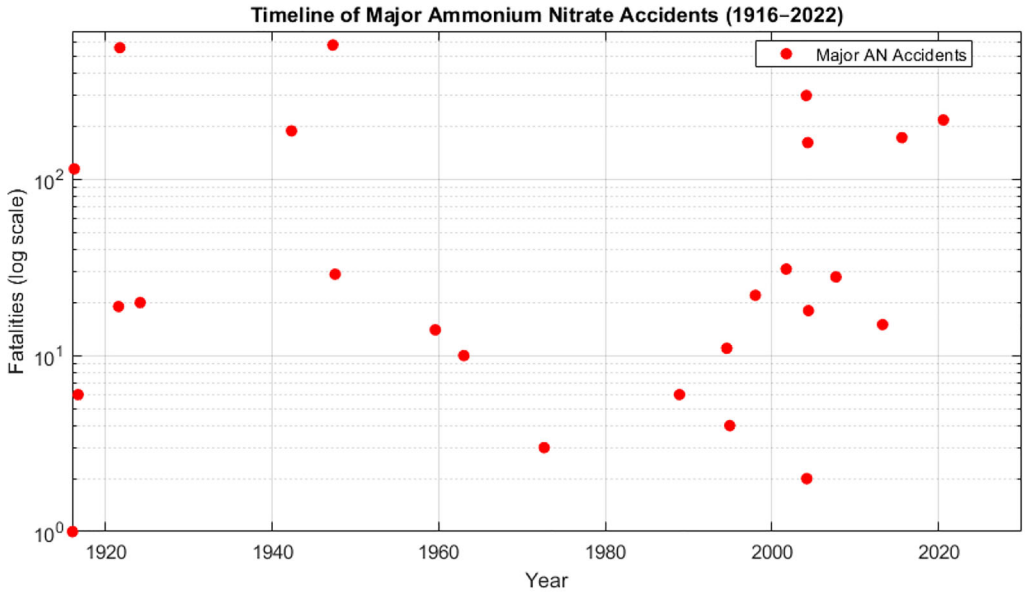

FIGURE 1 | The historical record of major ammonium nitrate accidents from 1916–2022. Fatalities are plotted on log scale for non-zero fatalities. Data are compiled from CAS (2021).

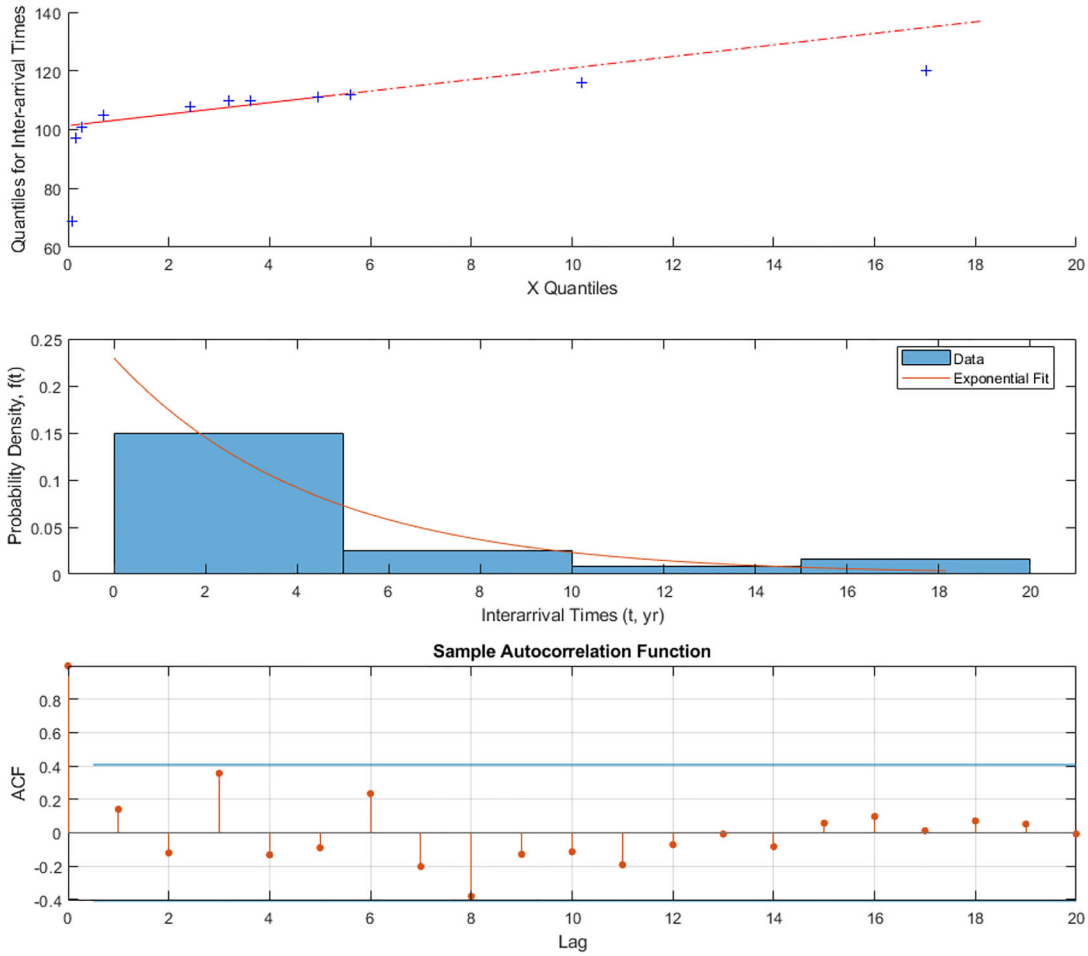

FIGURE 2 | Temporal analysis of AN accidents. (a) Exponential fit to inter-arrival times of 25 fatal accidents (24 intervals, $\lambda \approx 0.23$/year). (b) QQ plot confirming exponential distribution. (c) ACF plot showing zero autocorrelation (ACF ≈ 0). Data from 39 events (1916–2022), with 25 fatal accidents (casualties 1–581) driving hazard stability.

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TABLE 1 | Statistical comparison of fitted distributions for casualty data ($n = 25$).

Distribution	Parameters	KS Test (p-value)	AD Test (p-value)	AIC	BIC
Normal	$\mu = 101.5, \sigma = 163.6$	0.003	0.016	328.81	331.25
Lognormal	$\mu = 3.31, \sigma = 1.77$	0.550	0.801	268.17	270.61
GPD	$\xi = 1.13, \sigma = 24.09$	0.728	0.804	269.68	272.12

Note: Lower AIC/BIC values indicate a better relative model. p-values > 0.05 indicate the data do not significantly deviate from the tested distribution.

systems, not necessarily controlled improvements (Perrow 1984; Taleb 2020).

As temporal analysis indicated the events are not correlated, next we focus on modeling severity separately using a fat-tailed distribution to reveal the full scope of AN risk, unmitigated by historical "lessons learned."

2.2 | Heavy-Tailed Severity of AN Accidents

The severity of AN accidents, derived from 25 fatal events between 1916 and 2022 with casualties ranging from 1 to 581 (mean 101.5 > median 20), exhibits a heavy-tailed distribution. This significant "mean-median gap" (mean over five times the median) is a classic diagnostic for heavy-tailed phenomena, as the mean is heavily influenced by rare extremes (e.g., Texas City with 581 deaths), justifying EVT over thin-tailed Gaussian models (Taleb 2020). Graphical diagnostics, parametric modeling, and statistical hypothesis testing (Table 1) collectively confirm this pattern, challenging the thin-tailed assumptions often found in qualitative narratives like those from CAS (2021).

Before fitting any formal model, we study raw characteristics of the data. Here, graphical methods provide compelling evidence of heavy-tailed behavior. The Hill estimator plot (Figure 3a) shows the tail index α stabilizing around 1 for order statistics $k = 5$–15k, corresponding to a shape parameter $\xi \approx 1$. This indicates a potential for infinite mean, although small-sample variability ($n = 25$) is a noted limitation of the Hill estimator, which is known to be affected by variance–bias tradeoff in finite samples (Embrechts et al. 1997; Corral 2021). The variability due to small sample size is further checked via bootstrap sampling (10,000), computing the Hill estimator for each resample with a fixed threshold $u = 10$ casualties. The results, presented in Supplementary Figure S3, revealed a mean tail index $\alpha \approx 0.6158 \pm 0.1336$ and a mean shape parameter $\xi \approx 1.6917 \pm 0.3284$, aligning with mid-range Hill estimates and supporting potential heavy-tailed behavior ($\xi > 1$). The max-to-sum ratio plot (Figure 3b) demonstrates slow convergence for the first moment, reaching approximately 0.2 at $n = 25$, while higher moments (0.4–0.6) remain elevated, pointing to infinite variance as described by Taleb (2020). The mean excess plot (Figure 3c) shows a linear increase for thresholds $u < 50$, peaking around $u \approx 50$ with a mean excess of about 250. This positive slope provides empirical support for a positive shape parameter, $\xi > 0$, a feature of heavy-tailed behavior. The variability of the mean excess function increases beyond $u \approx 200$ due to sparse data points. These observations align with the diagnostic framework of Cirillo and Taleb (2016), where the trend of the mean excess function is used to infer fat-tailed phenomena in bounded data.

We now employ parametric modeling to further refine these observations. To this end, we have used maximum likelihood estimation (MLE) to fit two distributions commonly tested to describe such heavy tail behavior, namely, a generalized Pareto distribution (GPD) and a lognormal distribution, in addition to a normal distribution to serve as a baseline. The results are shown in Figure 4, and the parameter estimates are provided in Table 1. Indeed, Figure 4 illustrates that both GPD and lognormal cumulative distribution functions (CDFs) closely match the empirical data across the range of casualties, with the log-scale inset showing comparable tail behavior for both models, in contrast to the poorly fitting normal distribution. Moreover, Table 1 also provides test results for statistical validation: the Anderson–Darling (AD) test rejects the normal distribution ($p = 0.016$) but accepts both GPD ($p = 0.804$) and lognormal ($p = 0.801$), with Kolmogorov–Smirnov (KS) p-values exceeding 0.5 for the heavy-tailed models. Furthermore, in view of model selection, the Akaike information criterion (AIC) and Bayesian information criterion (BIC) show only a slight favor toward the lognormal model (268.17 vs. 269.68 for AIC), suggesting both fits are equally robust. This difference is tested statistically using Vuong's test, which returned a p-value of 0.125, indicating no significant preference between GPD and lognormal. Given the small sample size, the precise form of the distribution for describing the extremes cannot be conclusively determined at this stage, yet the data clearly exhibit heavy-tailed behavior.

In summary, the parametric analysis confirms that the distribution of AN accident casualties is not thin-tailed. Both the GPD and lognormal distribution provide statistically robust descriptions of the historical data, with no significant evidence favoring one parametric form over the other. This sets the stage for exploring the implications of these distributions in the subsequent societal risk assessment, which reveals distinct risk profiles.

2.3 | The Societal Risk Profile Is Intolerable

The societal risk profile of historical AN accidents was quantified using the frequency–number (F/N) curve, as presented in Figure 5. The empirical data, represented by black circles, reflect the observed safety performance across the 106-year study period (1916–2022) based on 25 fatal events with casualties ranging from 1 to 581 (mean 101.5). These data points fall within the "intolerable" risk region as defined by established criteria from the United Kingdom, Hong Kong, and

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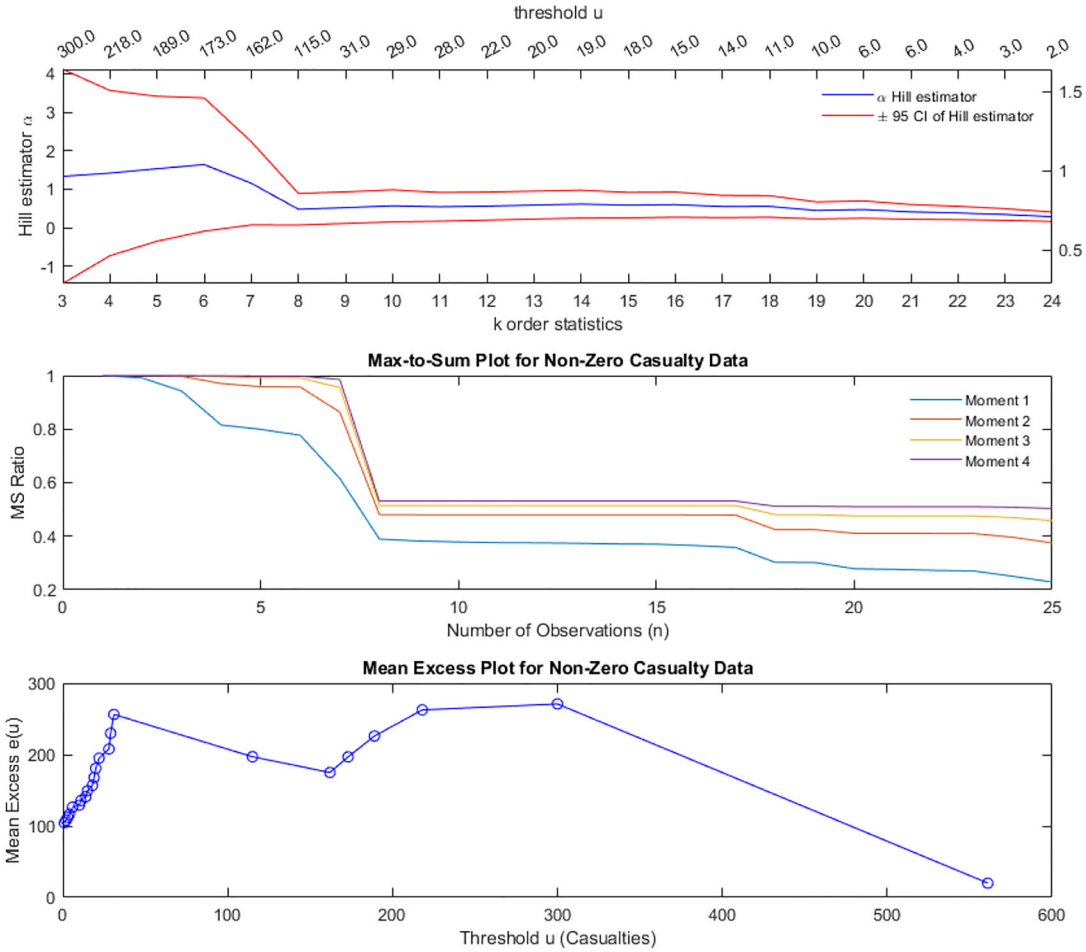

FIGURE 3 | Diagnostic plots for AN accident severity. Top: Hill estimator of tail index as a function of varying thresholds u / k upper order statistics; middle: max-to-sum ratio plot for moments $p = 1$ to 4 versus number of observations n; bottom: mean excess plot showing the mean excess $e(u)$ versus thresholds u.

the Netherlands, which set thresholds such as $F(N \geq 50) < 10^{-4}$ per year. For example, the observed frequency for $N \geq 100$ is approximately 0.094/year (10 events over 106 years period), which exceeds these regulatory benchmarks by 1–2 orders of magnitude.

To extrapolate this risk beyond the historical record, modeled F/N curves were derived by integrating the event rate λ = 0.229 events/year—determined by MLE of the Poisson process in Section 2.1—with the survival functions of the fitted GPD and lognormal distribution, as described in Section 2.2. Figure 5 illustrates that both modeled curves closely track the empirical data across the observed casualty range. However, their extrapolations for high-consequence, low-frequency events (e.g., $N > 581$, such as the Texas City incident) diverge: the GPD predicts a higher frequency of extreme events (e.g., $F(N \geq 1{,}000) \approx 0.002$/year) compared to the lognormal ($F(N \geq 1{,}000) \approx 0.0015$/year), reflecting the GPD's heavier tail. While both distributions fit the sample data well statistically (Vuong's test $p = 0.125$), this difference in tail projection is critical for risk inference. Adopting the precautionary principle, the GPD's conservative estimate is recommended for safety assessments, ensuring preparedness for rare but catastrophic outcomes.

We explore further the implications of this significant gap between the statistically characterized historical risk profile and modern safety benchmarks in the Discussion section.

3 | Discussion

The results of this study provide empirical evidence that the risk associated with AN accidents has not declined over time, challenging the prevailing narrative in process safety literature that suggests improvements in regulations and technology have reduced the likelihood and severity of such events. By applying EVT to a historical dataset of major AN accidents from 1916 to 2022, with a focus on the 25 fatal events, this analysis reveals a constant hazard rate and heavy-tailed casualty distribution, emphasizing the persistent and catastrophic nature of AN risks.

3.1 | Synthesis of the Findings: Persistent Hazard, Underestimated Tails, and Intolerable Societal Risk

Modern process safety assumes that risk is progressively reduced by regulation and technology. This is not supported by the

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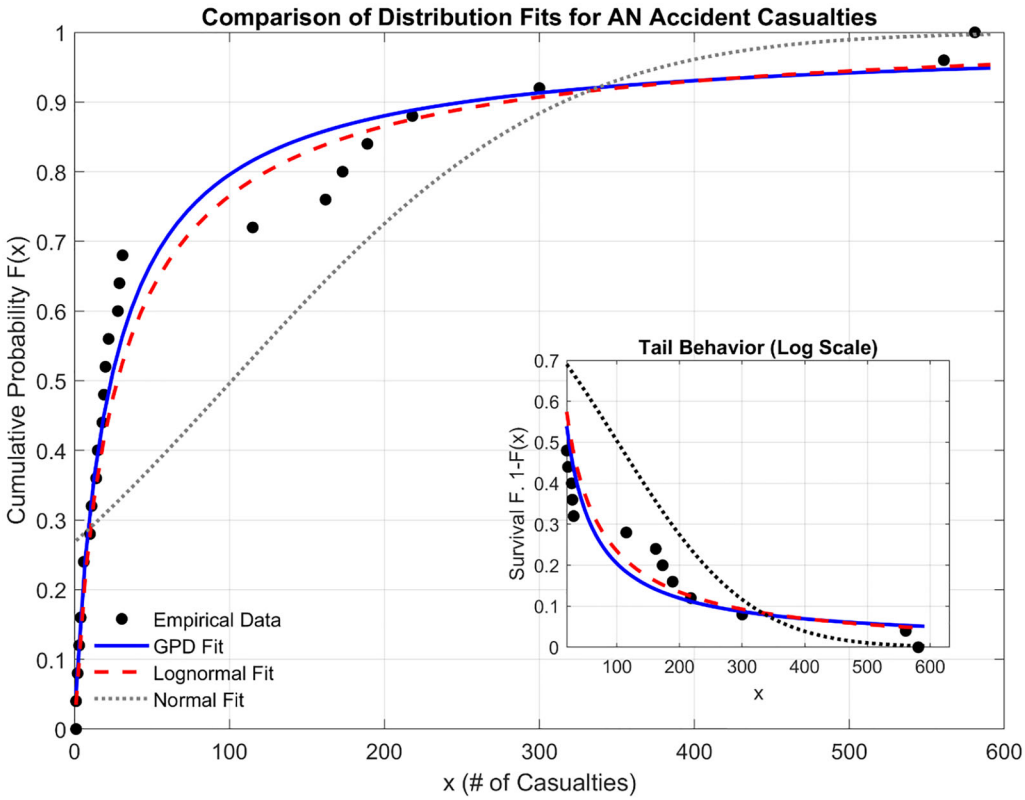

FIGURE 4 | Comparison of the empirical CDF of casualty data (circles) with the fitted CDFs for the GPD, lognormal, and normal distributions. Both heavy-tailed models show a strong visual fit, while the normal distribution deviates significantly.

historical record of AN accidents. Our analysis shows the Poisson hazard rate ($\lambda \approx 0.23$/year) has remained statistically constant for over a century, a finding that directly rebuke claims of improving safety. The presence of clusters and gaps in the accident events is consistent with a Poisson variance rather than evidence of progress in process safety. In this regard, long calm periods is a signal of stochastic luck rather than effective control of safety. This stability of the hazard rate means that such periods are natural fluctuations within a process. It contrasts the illusion of control argued by narrative-based approaches, which mistake the randomness of a stochastic process for a trend of improvement.

Complementing the temporal stability of hazard rate, we find that the casualty severity from 25 fatal AN accidents (1916–2022) follows a heavy-tailed distribution, as supported by both GPD ($\xi \approx 1.13$) and lognormal fits. This pattern qualifies catastrophic events—such as those exceeding 500 deaths—as systemic features rather than anomalies, despite ambiguity in the precise tail form due to the small sample size ($n = 25$). A key implication is the systematic underestimation of risk by conventional metrics. The sample mean of 101.5 deaths from the AN dataset suggests a manageable hazard, which aligns with narratives perceiving risks as controlled by modern measures (e.g., CAS 2021) or preventable (Babrauskas 2016). In contrast, shadow means—calculated via truncated GPD (physical bounds) and dual transformation (tail-sensitive; Cirillo and Taleb 2016)—reveal far greater potential (Table 2). For a Bhopal-scale bound (10,000 deaths cap), the truncated GPD mean is 1.5 times the sample mean, rising to 14 times at the global population scale (8.19 billion). The dual transformation yields even higher estimates of 1.7× and 19× of the sample mean, respectively. This disparity between the naïve sample mean of the dataset and the shadow mean (which is estimated using the fitted GPD distribution of the severity data) stems from the heavy-tailed nature of severity, where the law of large numbers converges too slowly to make the sample mean a reliable representative of the true distribution (Taleb 2020; Sin 2024). In other words, one accident event with high casualties in the future is enough to shift the sample mean significantly. Combined with the constant hazard rate, these findings quantitatively reframe AN risks as a persistent systemic challenge with broad implications for safety paradigms.

F/N curves, widely used by government agencies to regulate societal risks in chemical process safety (e.g., population exposures from major accidents), provide a clear benchmark for AN hazards (Figure 5). Both the empirical profile from 25 non-zero events ($F(N \geq 100) \approx 0.094$/year) and GPD extrapolations fall under the "intolerable" region under criteria from the UK, HK, and the Netherlands (thresholds $<10^{-4}$/year). The constant hazard rate thins to $\approx$0.05/year for severe thresholds but shows no decay over time. This leads us to conclude that AN risks persist as societally unacceptable at the aggregate level. Site-specific QRAs remain structurally unable to capture this ensemble-level reality, as they aggregate neither historical data nor nonlinear escalations that are computationally irreducible.

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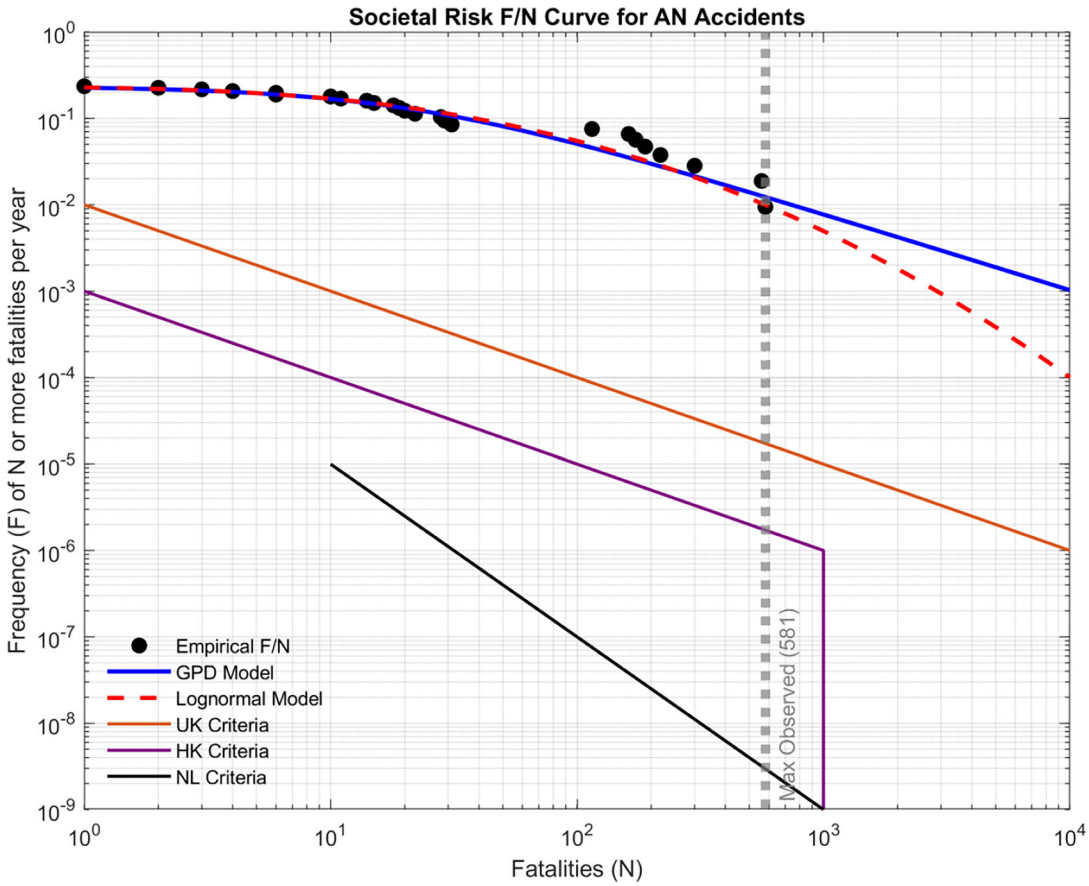

FIGURE 5 | Societal risk F/N curve for major AN accidents. The plot shows the empirical frequency (black circles) alongside the GPD (solid blue line) and lognormal (dashed red line) modeled frequencies. The models use the MLE-derived event rate (lambda = 0.229 events/year). Standard-risk acceptability criteria for the United Kingdom (UK), Hong Kong (HK), and the Netherlands (NL) are overlaid for comparison.

TABLE 2 | Shadow means derived from the AN dataset, contrasting truncated GPD (conservative estimate) and dual transformation (tail-sensitive estimate) against the sample mean ($n = 25$, $\xi \approx 1.13$), highlighting underestimation of casualty risk.

Bound	Sample mean	Truncated GPD mean	Dual transformation mean	Amplification (truncated/dual)
Bhopal (10,000)	101.52	148.65	173.60	1.5x / 1.7x
World (8.19 billion)	101.52	1431.59	1946.02	14x / 19x

3.2 | Theoretical Context: Complexity Science and Reductionist Limits

These findings show that AN accidents conform to patterns observed in other major catastrophes, with casualty severity following a heavy-tailed distribution akin to those in natural disasters and industrial events, such as earthquakes, hurricanes, and large-scale explosions (Embrechts et al. 1997). Therefore, conventional methods like sample mean, prove inadequate for analyzing properties of such severity data, as they rely on thin-tailed assumptions that overlook tail extremes.

These statistical features—a constant hazard rate and heavy-tailed severity—support theories explaining failures in complex systems. In particular, the data align with normal accident theory (Perrow 1984), which posits that "major incidents" are an emerging and inevitable outcome of tightly coupled systems like the AN supply chain. This point of view is further supported by the computational irreducibility concept of Wolfram (2017). In short, it says that specific failure pathways remain fundamentally unpredictable by using shortcut models like QRA models. The observed persistence and tails thus emerge as macroscopic manifestations of underlying irreducible complexities, including latent organizational and political factors such as regulatory lapses, economic pressures, and complacency cultures. QRAs cannot formally incorporate these factors through predefined models. The 2020 Beirut disaster is an example of this point, where governance failures transformed a localized hazard into 218 fatalities and a $15B economic catastrophe. That is why we argue here that one needs a statistical approach, such as our EVT-based

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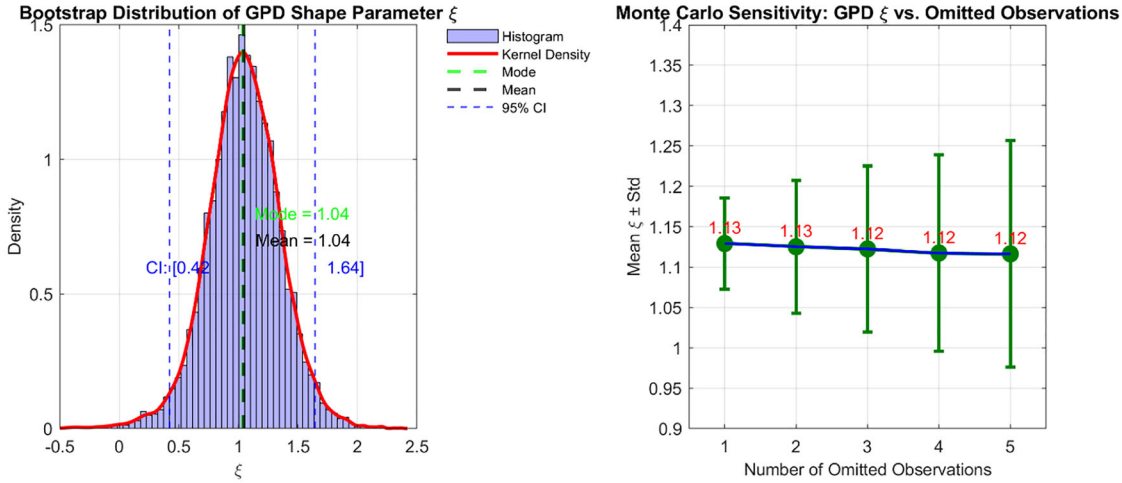

FIGURE 6 | Robustness and sensitivity check due to data uncertainty and limitations: left: bootstrap sampling from the data and calculation of the shape parameter; right: Monte Carlo sensitivity analysis of data omissions showing a stable estimate for the shape parameter up to 20% of omissions.

ensemble framework, to assess these realized systemic effects from historical data. This we posit will complement site-specific approaches for holistic governance.

3.2.1 | Robustness and Data Limitations

The validity of these conclusions was rigorously tested. The stability of the GPD shape parameter, ξ, was confirmed via bootstrap resampling (Figure 6, left), which produced a 95% confidence interval of [0.42, 1.64]. The interval is entirely above zero, confirming that the heavy-tailed finding is robust to sampling variability. A parallel bootstrap analysis of the lognormal parameters likewise confirmed their stability.

Furthermore, a Monte Carlo sensitivity analysis addressed potential data incompleteness. The estimate for ξ remained stable around 1.12 even with 20% of the data randomly omitted (Figure 6, right), demonstrating that the conclusions are not artifacts of a few influential events. The primary limitation of this study is the small sample size ($n = 25$) for severity analysis, which is the source of the ambiguity between the GPD and lognormal models. However, despite this limitation, the data are sufficiently robust to support the study's two central conclusions: the hazard rate is constant, and the severity distribution is clearly heavy-tailed. Furthermore, other attributes of these accidents, such as environmental and economic impacts/losses, can also be studied to further analyze the tail behavior of these accidents.

3.3 | Practical Applications: Rethinking Safety Management With a Systemic Framework

The robust patterns revealed in this study highlight a persistent and intolerable risk, calling for a rethinking of AN safety management philosophy—from a site-specific, reactive approach to a holistic probabilistic one. Current frameworks, such as the CCPS risk-based process safety model (CCPS, 2007) and regulatory guidance from agencies like the EPA (2021) or UK HSE, focus on managing hazards at individual facilities but undervalue the presence of systemic heavy tails across the industrial ensemble.

Therefore, we propose to shift the focus from site-specific monitoring to a more holistic level with an EVT-based systemic monitoring framework. By leveraging the historical ensemble, this approach treats the global AN supply chain as a complex, tightly coupled system where events with catastrophic casualties scale with time. This approach is aligned with managing extreme event risk in many contemporary domains, from hydrological engineering (flood risks) to financial, insurance (Embrechts et al. 1997), and aviation risks (Das and Dey 2016).

Using the current baseline risk climate derived from our modeling ($\lambda \approx 0.23$ events/year; $\xi \approx 1.13$), we establish reference benchmarks to monitor systemic safety performance (Figure 7; Table 3).

Based on the risk climate characterized by this study (1916–2022), the model projects a 50-year return level ($T = 50$) of approximately 315 fatalities and a 100-year return level ($T = 100$) of approximately 715 fatalities (Table 3). These benchmarks provide a transition from tracking arbitrary industry thresholds (e.g., accidents with fatalities >30; CAS 2021) to monitoring the actually observed "return rate" of extreme events on a system level.

It is important to emphasize the stochastic nature of these accidents for a proper interpretation of Table 3. In the context of the AN supply chain, a 50-year return level of 315 fatalities does not mean that an accident will happen exactly every 50 years, nor does it mean the system is safe from accidents for 50 years after such an accident occurs. Because these accidents are stochastic in nature (randomly occurring in time and place) and follow a memoryless Poisson process, the return period describes the average frequency of events with such magnitudes, not a deterministic and fixed arrival date.

The exposure probability is used here to inform us about the likelihood of an event occurring within a specific risk-monitoring

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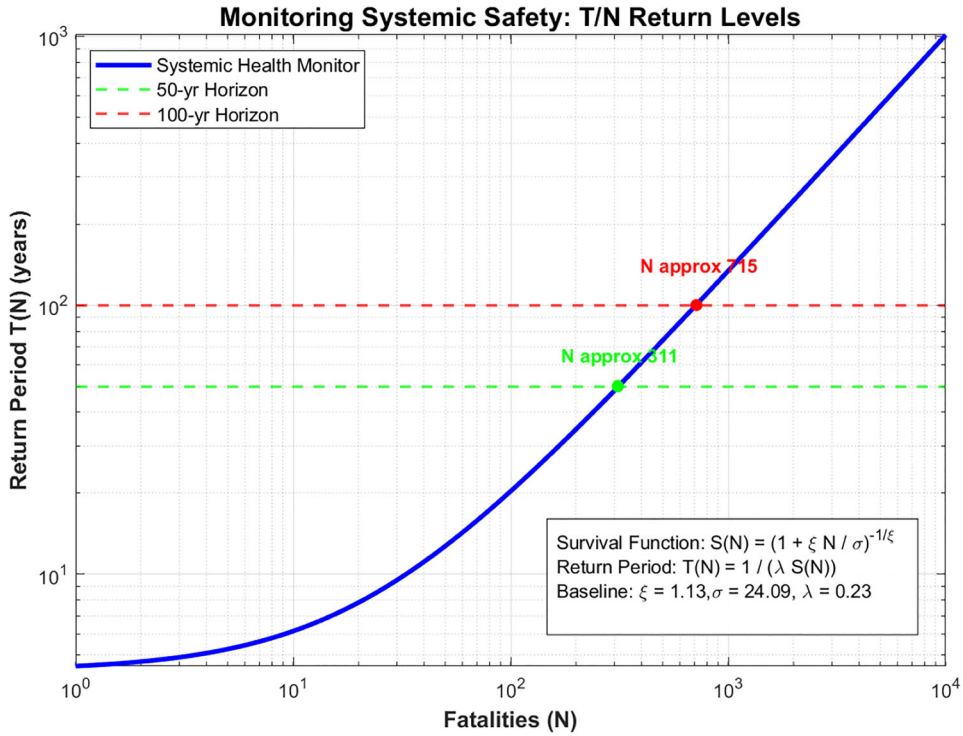

FIGURE 7 | Systemic monitoring T/N curve showing return levels for major accidents in the global AN supply chain. The observed near-linear slope on log–log scale, especially for N larger than 100, illustrates the scaling relationship between rarity (return period, T) and severity (magnitude, N) in a heavy-tailed regime.

TABLE 3 | Systemic monitoring benchmarks (based on the risk climate characterized by 1916–2022).

Benchmark event ($T(N)$)	Return level (N) fatalities)	Annual probability ($H = 1$)	50-Year horizon prob. ($H = 50$)	100-Year horizon prob. ($H = 100$)
50-Year event	~315	~2.0%	~63%	~86%
100-Year event	~715	~1.0%	~39%	~63%

Note: The exposure probability, $P(N, H)$, is the likelihood of observing at least one accident exceeding N fatalities within a given monitoring horizon H. In a Poisson process, it is calculated as $P\ (N, H) = 1 - \exp(-F(N) \times H)$, which simplifies to the intuitive form, $P = 1 - \exp(-H/T)$, where $T = \frac{1}{F(N)}$, is the return period. For any horizon $H = T$, the probability of occurrence is approximately $P = 1 - \exp(-1) \approx 63.2\%$.

window. For example, there is an approximately 63% chance that an event with a magnitude of 315 fatalities will occur within a 50-year time horizon ($H = 50$). On the other hand, this also means that there is still a 37% chance that such an event will not occur during that same period.

This framework provides a robust method for evaluating the effectiveness of global safety interventions and technological advances cumulatively. Within a stationary risk climate characterized by a homogenous Poisson process, the annual frequency (F(N))—and consequently the annual probability—of a major event remains constant and is independent of when the last major accident, such as Beirut (Al-Hajj et al. 2021), occurred. For systemic monitoring, the primary focus is on an increase in the return period (T). If regulatory reforms post-Beirut were truly impactful on a system level, we would expect to observe the return period for a 315-fatality event increase significantly beyond its current 50-year baseline. Such an increase in T would mathematically imply a reduction in the annual hazard rate, thereby lowering the annual exposure probability.

However, so far, the ensemble statistics demonstrate that the average inter-arrival times for major accidents have remained largely constant over the past century. This suggests that the underlying systemic risk profile of the global supply chain has not been fundamentally changed, despite the progress of technical guidelines and localized safety protocols. In short, this problem is not solved, and demands continued focus and vigilance from all stakeholders involved in this complex and critical AN supply chain. Until the frequency of these high-consequence events is reduced at a systemic level, these risks must be treated as persistent features of the global industrial landscape rather than isolated anomalies.

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4 | Conclusions and Perspectives

We empirically validate the hypothesis from the introduction: AN accidents exhibit persistent heavy-tailed distributions, driven by a constant hazard rate and severe tails that position extreme casualties (>500 deaths) as systemic features rather than anomalies. Applying EVT to the historical ensemble of 39 major accidents (1916–2022; 25 fatal events), we model temporal occurrence as a homogeneous Poisson process ($\lambda \approx 0.23$/year) and severity as heavy-tailed (GPD $\xi \approx 1.13$). These findings indicate that systemic risk patterns have remained unchanged for over a century.

We propose, therefore, an EVT-based systemic monitoring framework as a complementary method to current practice. This framework quantifies realized systemic risks from historical data, enabling the establishment of quantitative benchmarks—such as the 50-year (~315 deaths) and 100-year (~715 deaths) return levels—over mechanism-driven simulations or narratives. This shift can be extended beyond AN to other chemicals in broader process industries and calls for international data-sharing for studying the true stochastic nature of chemical accidents. Future work could incorporate economic as well as environmental impacts to expand tail estimates to other impacts of the accidents, advancing chemical safety toward a data-driven foundation.

5 | Materials and Methods

5.1 | Dataset and Uncertainty

The dataset consists of 39 major AN accidents recorded between 1916 and 2022, compiled from public literature sources such as CAS (2021) primarily and cross-checked with Pittman et al. (2014) and Babrauskas (2016). Each entry includes the date, location, and estimated number of casualties detailed in Supplementary Table S1. Of these, 25 events recorded non-zero casualties, ranging from 1 to 581 deaths, forming the full sample for distribution analysis. The dataset is subject to uncertainties, including potential under-reporting of minor incidents and imprecise casualty estimates, often based on historical records, eyewitness accounts, or post-event investigations (CAS 2021). To mitigate these, robustness was assessed using bootstrapping (resampling with replacement to estimate parameter variability) and Monte Carlo resampling techniques, simulating scenarios with up to 20% of data missing or inaccurate (Cirillo and Taleb 2016). All statistical analyses were conducted using MATLAB's Statistics Toolbox (version R2023a or later), with MLE implemented via the fitdist function. Custom MATLAB scripts were developed for graphical diagnostics (e.g., mean excess plots) and the Hill estimator, ensuring reproducibility.

5.2 | Temporal Analysis: Testing for Constant Hazard

To assess whether the hazard rate of AN accidents is constant over time, we analyzed the temporal pattern of accidents by examining inter-arrival times, *Ti*, defined as the time differences (in years) between consecutive accident dates. The dataset comprises 39 recorded AN accidents from 1916 to 2022 (Supplementary Table S1), of which 25 had non-zero casualties (1–581 deaths), yielding 24 inter-arrival times for the temporal analysis of fatal events ($n = 25$).

A homogeneous Poisson process assumes that accidents occur randomly and independently at a constant average rate λ (Embrechts et al. 1997). Under this model, the inter-arrival times T_i follow an exponential distribution with probability density function (PDF), $f(T) = \lambda e^{-\lambda T} for T \geq 0$, where $\lambda > 0$ is the rate parameter, and CDF, $F(T) = 1 - e^{-\lambda T}$. The mean inter-arrival time is $1/\lambda$, and the process is called memoryless, which means that the probability of an event in the next interval is independent of previous occurrences.

To test this hypothesis, QQ plots were constructed by comparing the empirical quantiles of the observed inter-arrival times against the theoretical quantiles of an exponential distribution. In a typical interpretation, a close alignment to the 45-degree line in QQ plot supports the exponential hypothesis. The autocorrelation function (ACF) is used to check the presence or absence of a memoryless process: $\rho_k = \frac{Cov(T_i, T_{i+k})}{Var(T_i)}$, where *Cov* denotes covariance and *Var* variance; ρ_k should be close to 0 for all lags k > 0, in which case it would be indicating absence of a temporal correlation.

The rate parameter λ was estimated using MLE via MATLAB's fitdist function on inter-arrival times of 25 non-zero casualty events (24 inter-arrivals) with 95% confidence intervals derived from the asymptotic covariance matrix. A dispersion test, based on the variance-to-mean ratio of event counts per decade, assessed over-dispersion, with a p-value > 0.05 supporting the Poisson assumption. To examine the thinning of λ with increasing casualty thresholds, a POT analysis was performed, as higher-severity events may occur at a reduced rate (Embrechts et al. 1997). Bootstrap resampling (B = 10,000) provided additional sensitivity checks for robustness (Supplementary S2).

Having modeled the timing of fatal AN accidents, we next analyze their severity to quantify their catastrophic potential.

5.3 | Severity Analysis: Characterizing the Full Distribution

5.3.1 | Graphical Diagnostics for Tail Behavior

To characterize the distribution of casualty data X_i ($i = 1, \ldots, n = 25$ non-zero values), we first employ graphical diagnostic methods to assess tail behavior and test for thin-tailed vs. heavy-tailed properties.

The mean excess function (mef), following Cirillo and Taleb (2016), is defined as the expected excess over a threshold u, $e(u) = E(X - u | X > u)$, which is calculated empirically for a sample as follows:

$$e_n(u) = \frac{\sum_i^n (X_i - u)}{\sum_i^n 1_{\{X_i > u\}}}.$$

Here, threshold u is defined as $0 < u < x_f$ and $x_f = \max(x, 0)$ is the supremum point in the sample, $1_{\{condition\}}$ is the indicator

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function. The mef is commonly used for identifying heavy tails, which exhibits a positive linear slope with u for a sample drawn from distributions with a finite tail index, indicating that the average exceedance grows systematically with the threshold (Cirillo and Taleb 2016).

Maximum-to-sum plot is another graphical method useful for studying the presence of paretinaity in data. This plot is used to analyze the contribution of maximum values in a sample to a given moment. For a given sample of random variates $\{X_1, X_2 \ldots X_n\}$, this ratio is computed as follows: $R_n^p = \frac{M_n^p}{S_n^p}$. Here $S_n^p = \sum_{i=1}^{n} X_i^p$ is the partial sum and $M_n^p = \max(X_1^p \ldots X_n^p)$ is the partial maximum for a given p moment. For thin-tailed distributions (e.g., normal), R_n^p converges to 0 as m increases for all finite p, reflecting finite moments. For heavy-tailed distributions, R_n^p converges slowly or not at all for p exceeding the tail index, suggesting infinitely higher-order moments (Taleb 2020).

In addition, the Hill estimator, a non-parametric method, is used to study shape parameter ξ (inverse of the tail index, $\alpha : \xi = \alpha^{-1}$) and provide a measure of the heaviness of the tail in the data. For a given sequence of random variables, X_j and threshold u,

$$\widehat{\alpha_k} = \left(\frac{1}{k} \sum_{j}^{k} lnX_{j,k} - lnu \right)^{-1},$$

where $X_{j,k}$ are the k upper order statistics exceeding a threshold u, and u is chosen from a range of thresholds to evaluate stability. The estimator is then plotted against k or u to identify a plateau, indicating whether a consistent tail index can be inferred (Embrechts et al. 1997). We use the Hill estimator as a diagnostic step in this study, before we perform a parametric fitting.

5.3.2 | Distribution Fitting and Sensitivity Check

The full sample of 25 non-zero casualties was fitted with two heavy-tailed distributions using MLE to comprehensively describe the data. This full-sample approach was chosen over threshold-based methods like POT (Embrechts et al. 1997) due to the limited sample size. Otherwise, in the POT approach, it would be reduced to an unacceptably small number of exceedances (e.g., sample size is 17 at threshold, $u = 10$). This approach is aligned with Corral (2021), who also argued that full distributions are sufficient for sparse datasets, especially if they describe the data well. However, as a sensitivity check, we have also done and presented the POT-based approach for describing the tail of the data, which is given in the supplementary material.

The fitted distributions were:

- GPD: With location parameter $\mu = 0$ (Pareto Type II or Lomax distribution), suitable for positive-support data ($X > 0$). The PDF is:
  $f(X;\ \xi, \sigma) = \frac{1}{\sigma}(1 + \xi \frac{X}{\sigma})^{-(1/\xi+1}$ for $X > 0, \xi > 0$, and $\sigma > 0$, where ξ is the shape parameter governing tail heaviness, and σ is the scale parameter. The CDF is:
  $F\ (X;\ \xi, \sigma) = 1 - (1 + \xi \frac{X}{\sigma})^{-(1/\xi}$. The interpretation of the shape parameter is as follows: for $\xi > 0$, the distribution is considered heavy-tailed; $\xi \geq 1$ implies an infinite mean, and $\xi \geq 0.5$ an infinite variance. GPD parameters were estimated via MLE using MATLAB's fitdist, with awareness of potential convergence issues (Grimshaw 1993) and small-sample bias (Giles et al. 2016). Given our estimated $\xi \approx 1.13 > 1/3$, the analytic $O(1/n)$ correction in Giles et al. (2016) is not directly applicable under their regularity conditions (valid only for $-1/3 < \xi < 1$).
- Lognormal distribution: A heavy-tailed distribution (depending on its parameters) with finite moments, defined by the PDF as follows:
  $f\ (X; \mu, \sigma) = \frac{1}{X\sigma\sqrt{2\pi}} e^{-\frac{(lnX-\mu)^2}{2\sigma^2}}$ for $X > 0$where μ is the mean of the natural logarithm of the variable, and σ is the standard deviation on the log scale. The CDF is given as:
  $F(X; \mu, \sigma) = \frac{1}{2}(1 + erf\{\frac{(\ln X-\mu)}{\sigma\sqrt{2}}\})$,where *erf* is the error function. MLE was implemented via fitdist (data, 'Lognormal') in MATLAB.

For a baseline comparison, a normal distribution was fitted: PDF $f(X; \mu, \sigma) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(X-\mu)^2}{2\sigma^2}}$, where μ is the mean and σ^2 the variance, and CDF $F(X; \mu, \sigma) = \frac{1}{2}(1 + erf\{\frac{(X-\mu)}{\sigma\ \sqrt{2}}\})$. This function was used for testing the presence of the thin-tailed hypothesis in the dataset.

5.3.3 | Dual Distribution for Bounded Tails and Shadow Mean

Working with power law distributions such as GPD may indicate the presence of infinite moments depending on the shape parameter. However, we are dealing with modeling casualties X, which is constrained between a lower limit $L \approx 0+$ *(approximating the minimum observed casualty of 1)* to plausible upper limits H (e.g., 10^4 for a Bhopal-scale maximum or 8.19 billion for world population as considered in Cirillo and Taleb 2016). Therefore, we used two approaches to address potential infinite-mean biases in the heavy-tailed fits. First, an intuitive approach is used where a truncated GPD mean was calculated as the expected value over a finite interval (Corral 2021):

$$E\ \ (X|0 \geq X \geq U) = \int_0^U x f\,(x; \xi, \sigma)\, dx / F\,(U)\,,$$

where $f(x)$is the GPD pdf and $F(U)$is CDF, fitted to the 25 non-zero casualty events. This provided a conservative estimate by capping the tail at U.

Second, we used the dual transformation concept of Cirillo and Taleb (2016) to refine tail estimation. Here we define the dual variable: $Z = -H\ ln(1\ -\ X/H)$ to transform the bounded X into an unbounded Z space, with the inverse $X = H\ (1 - e^{(-Z/H)})$. The shadow mean, a finite proxy for the distribution's tail behavior (in power law distributions, mean will be closer to the tail), was estimated as follows: a GPD was fitted to the transformed sample Z_i, with parameters ξ_Z (shape) and σ_Z (scale), reflecting the tail structure of the original data under the transformation. The expected value $E(X)$ was then computed using Monte Carlo simulation as follows: we generated 10^6 samples from $Z^* \sim$

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$GPD(\xi_z, \sigma_z)$ and then used the inverse transformation, $X^* = H\,(1 - e^{(-Z^*/H)})$, to obtain sample of X^*, and taking the sample mean as the shadow mean. The large sample size of Monte Carlo simulations is used to ensure high precision (standard error $\sim 1/\sqrt{10^6}$), aligning with empirical robustness for the dataset's small size ($n = 25$). Using H = 10^4 or 8.19×10^9 to align with the truncation bounds U, this method accounts for unobserved extremes near H, enhancing the analysis of the dataset's heavy-tailed structure ($\xi\approx1.13$).

5.3.4 | Model Comparison and Goodness-of-Fit

The quality of the fitted distributions was evaluated using several approaches. To compare the non-nested GPD and lognormal models, we employed Vuong's closeness test, a formal hypothesis test designed to determine if one model is significantly closer to the true, unknown data-generating process than the other. The test evaluates the null hypothesis that both models are equally close.

In addition, the AIC and BIC were calculated to measure the relative quality of each model. These criteria provide a trade-off between goodness-of-fit (likelihood) and model complexity (number of parameters), with lower values indicating a preferred model.

The absolute goodness-of-fit for each model was assessed using standard statistical tests. The KS test was used to compare the empirical CDF of the data against the fitted CDFs. The AD test, which gives more weight to the tails of the distribution, was also used to complement the KS test. A p-value greater than 0.05 for these tests indicates that the data are not inconsistent with the fitted model.

5.3.5 | Robustness and Sensitivity Analysis

The stability of the conclusions was assessed through methods whose results are presented in the Discussion section. The variability of the key parameters for the candidate heavy-tailed models (e.g., shape parameter ξ for the GPD) was estimated using bootstrap resampling (10,000 resamples with replacement). Furthermore, a Monte Carlo analysis was used to test the sensitivity of the results to potential data incompleteness by randomly omitting up to 20% of the observations and re-fitting the models. Now that we have completed our severity modeling, next we quantify societal risk using F/N curves.

5.4 | Societal Risk Quantification: F/N Curve

Societal risk was quantified using an F/N curve, which plots the cumulative frequency (F) of accidents resulting in N or more fatalities against the number of fatalities (N) on a log–log scale. The empirical F/N curve was constructed directly from historical data, where the frequency for a given number of fatalities N is calculated as $F(N) = k\,/\,T$, with k being the number of historical accidents with casualties greater than or equal to N, and T being the total observation period (106 years).

The modeled F/N curve, used for extrapolation and comparison, was derived by combining the results of the temporal and severity analyses. The frequency $F(N)$ is the product of the overall accident rate, λ (from the homogeneous Poisson process), and the survival function, $S(N)$, of the fitted severity distribution: $F(N) = \lambda \times S(N)$. For this study, $S(N)$ was calculated using the parameters of both the fitted GPD and lognormal distributions for comparison. These curves were then plotted against standard societal risk acceptability criteria from the UK, HK, and the Netherlands (NL) to assess the tolerability of the observed risk.

Return periods were calculated as the inverse of $F(N)$, $T(\mathrm{N}) = 1\,/\,F(N) = 1\,/\,(\lambda * S(N))$, providing the expected average time (years) between events exceeding N casualties (Embrechts et al. 1997). In the EVT framework, the occurrence of accidents follows a homogenous Poisson process with a constant rate ($\lambda = 0.23$ /year), the inter-arrival times of events that exceed magnitude N follow an exponential distribution with a mean of $T(N)$. Here $S(N)$ is the survival function derived from the fitted GPD. This approach provides the theoretical basis for the benchmarks established in Section 3.4, specifically the 50-year and 100-year return levels, which reflect the systemic hazard rate identified across the century-long historical ensemble.

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Data availability

The dataset consists of 39 major AN accidents (1916–2022), compiled from public literature sources such as CAS (2021), cross-checked with Pittman et al. (2014) and Babrauskas (2016). It includes details on date, location, and casualties and is provided in full in Supplementary Table S1 (S1: Accident Data Table). Custom MATLAB scripts for analyses (e.g., MLE via fitdist, bootstrapping, and Hill estimator) are available from the corresponding author upon reasonable request.

References

Al-Hajj, S., H. R. Dhaini, S. Mondello, H. Kaafarani, F. Kobeissy, and R. G. DePalma. 2021. "Beirut Ammonium Nitrate Blast: Analysis, Review, and Recommendations." *Frontiers in Public Health* 9: 657996. https://doi.org/10.3389/fpubh.2021.657996.

Babrauskas, V. 2016. "Explosions of Ammonium Nitrate Fertilizer in Storage or Transportation Are Preventable Accidents." *Journal of Hazardous Materials* 304: 134–149.

CAS. 2021. Ammonium Nitrate Explosions: Lessons Learned [White Paper]. American Chemical Society. https://cdn.prod.website-files.com/650861f00f97fe8153979335/66198f30486c46cf26d57653_CASGENENGWHP100044-Ammonium-Nitrate-White-Paper-Digital.pdf.

Center for Chemical Process Safety (CCPS). 2007. *Guidelines for Risk-Based Process Safety*. John Wiley & Sons, Inc. https://doi.org/10.1002/9780470925119.

Cirillo, P., and N. N. Taleb. 2016. "On the Statistical Properties and Tail Risk of Violent Conflicts." *Physica A: Statistical Mechanics and Its Applications* 452: 29–45. https://doi.org/10.1016/j.physa.2016.01.050.

Corral, Á. 2021. "Tail of the Distribution of Fatalities in Epidemics." *Physical Review E* 103: 022315. https://doi.org/10.1103/PhysRevE.103.022315.

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Das, K. P., and A. Dey. 2016. "Quantifying the Risk of Extreme Aviation Accidents." *Physica A: Statistical Mechanics and Its Applications* 463: 345–355. https://doi.org/10.1016/j.physa.2016.07.023.

Embrechts, P., C. Klüppelberg, and T. Mikosch. 1997. *Modelling Extremal Events for Insurance and Finance*. Springer.

Giles, D. E. A., H. Feng, and R. T. Godwin. 2016. "Bias-Corrected Maximum Likelihood Estimation of the Parameters of the Generalized Pareto Distribution." *Communications in Statistics—Theory and Methods* 45: 2465–2483.

Grimshaw, S. D. 1993. "Computing Maximum Likelihood Estimates for the Generalized Pareto Distribution." *Technometrics* 35: 185–191.

Perrow, C. 1984. *Normal Accidents: Living With High-Risk Technologies*. Princeton University Press.

Pittman, W., Z. Han, B. Harding, et al. 2014. "Lessons to Be Learned From an Analysis of Ammonium Nitrate Disasters in the Last 100 Years." *Journal of Hazardous Materials* 280: 472–477.

Sin, G. 2024. "Global Sensitivity Analysis Using Monte Carlo Estimation Under Fat-Tailed Distributions." *Chemical Engineering Science* 294: 120124. https://doi.org/10.1016/j.ces.2024.120124.

Sivaraman, S., and S. Varadharajan. 2021. "Investigative Consequence Analysis: A Case Study Research of Beirut Explosion Accident." *Journal of Loss prevention in the Process industries* 69: 104387.

Taleb, N. N. 2020. *Statistical Consequences of Fat Tails: Real World Preasymptotics, Epistemology, and Applications*. STEM Academic Press.

U.S. Environmental Protection Agency (EPA). 2021. "Chemical Advisory: Solid Ammonium Nitrate (AN) Storage, Handling and Management. EPA Office of Chemical Safety and Pollution Prevention." https://www.epa.gov/rmp/chemical-advisory-solid-ammonium-nitrate-storage-handling-and-management.

Wolfram, S. 2017. "A New Kind of Science: A 15-Year View." *Complex Systems* 1: 1–1197.

Supporting Information

Additional supporting information can be found online in the Supporting Information section.

Table 1S lists the raw data used in the major accidents analysis. Figure S2.1 Bootstrap estimation of hazard rate parameter λ Figure S2.2 Hazard rate parameter, λ, dependency for different threshold casualties. FigureS3.1 The bootstrap estimation of tail index and shape parameter using hill estimator. This is compared against the reference estimates obtained via GPD MLE fit to the data.

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